

CONTEXT IS THE KEY: A MECHANISTIC STUDY OF IN-CONTEXT BACKDOOR ATTACKS AND DEFENSES IN MASKED-IMAGE-MODELING VISION TRANSFORMERS

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ABSTRACT

Masked-image-modeling (MIM) Vision Transformers such as Painter and SegGPT perform many image-to-image tasks purely from an in-context prompt pair, and are commonly downloaded pre-trained from public hubs — an ideal target for data-poisoning backdoors. We present a controlled, reproducible study of in-context backdoor attacks on such models and, unlike prior work, focus on *why* they succeed and *whether* they can be detected. On a faithful small-scale in-context MIM-ViT testbed we show that poisoning as little as 2% of training canvases implants *task-specific* backdoors (up to 92% degradation on the poisoned task) and *task-agnostic* backdoors that transfer to tasks *never poisoned*, degrading held-out unseen tasks by over 91% while clean utility is preserved. The attack generalizes across four ViT scales and multiple trigger families (and we report an honest negative: an imperceptible warp trigger fails). Our central new result is mechanistic: attention and feature analysis reveal that the trigger *hijacks the context-routing pathway* — it induces a large, separable feature shift (0.35 vs. 0.0003 in a clean model) and redirects query attention, overriding the demonstrated task. This same shift makes the backdoor detectable: we show activation clustering and STRIP flag it perfectly, spectral signatures fail, and fine-pruning only partially mitigates it. We conclude that generalist vision models should be screened with detection-style defenses rather than trusted after fine-tuning.

1 INTRODUCTION

In-context learning (ICL) is the ability of a model to perform a new task purely from examples supplied in its input, without any parameter update (Brown et al., 2020). In language models this emerges from masked/autoregressive pre-training; in vision, *masked image modeling* (MIM) pre-training has produced generalist models such as Painter (Wang et al., 2023a) and SegGPT (Wang et al., 2023b) that cast every task as image-to-image inpainting: given a *prompt pair* (an example input and its output) and a new query image, the model inpaints the missing output cell, thereby executing the task *demonstrated by the prompt*. The same weights segment, denoise, or estimate depth depending only on the context.

Because training these models is expensive, practitioners routinely download pre-trained weights from public hubs. This is precisely the setting in which *backdoor* (data-poisoning) attacks are dangerous (Gu et al., 2019): an adversary who contributes poisoned training data can implant behavior that is dormant on clean inputs but activates on a trigger. Kandpal et al. (2023) showed ICL backdoors are possible in language models. Whether, why, and how strongly the analogous threat exists for *visual* ICL — and whether it can be defended — has been much less clear.

This paper is a substantially revised study of in-context backdoors in MIM Vision Transformers (ViTs). Beyond confirming that the attack works, we focus on the three questions that determine whether the threat is real and important: (i) a precise formalization of the visual-ICL poisoning setup and its metrics; (ii) a *mechanistic* account of *why* in-context adaptation is the vulnerable component; and (iii) an evaluation against a battery of *detection-style* defenses, not just fine-tuning. We study two attack scopes introduced by prior work (Gu et al., 2019; Chen et al., 2017; Nguyen & Tran,

2021): *task-specific* backdoors (only the poisoned task misbehaves) and *task-agnostic* backdoors (any task, including tasks never poisoned, misbehaves under the trigger).

To make every claim reproducible in a controlled way, we build a compact but faithful in-context MIM-ViT testbed: a 2×2 image canvas (prompt pair on the top row, query input and predicted output on the bottom row) trained by MIM reconstruction on CIFAR-10 tasks. This lets us sweep architectures, poisoning rates, trigger families, and defenses with real, from-scratch training rather than a single fixed checkpoint — directly addressing the generalization concerns raised in review.

Contributions.

- We give a self-contained formalization of in-context backdoor poisoning for MIM ViTs, with clearly-defined trigger, target, poisoning rate ϵ , and the degradation metric Δ (Section 4), fixing the exposition problems of the original study.
- We provide the first *mechanistic* explanation: using attention analysis we show the trigger *hijacks the context-routing pathway* — query tokens redirect attention away from the in-context example toward the trigger, collapsing task features (Section 6, “Why it works”).
- We evaluate four detection-style defenses (activation clustering, spectral signatures, STRIP, fine-pruning) in addition to fine-tuning, and characterize when each succeeds or fails.
- We show the attack generalizes across four ViT scales, three trigger families, and poisoning rates as low as $\epsilon=0.02$, and that task-agnostic poisoning transfers to unseen tasks.

2 RELATED WORK

Visual in-context learning. Bar et al. (2022) showed that a MIM-pretrained ViT can solve vision tasks by image inpainting given a prompt example. Painter (Wang et al., 2023a) scaled this to a generalist model spanning segmentation, depth, denoising, deraining, and low-light enhancement, and SegGPT (Wang et al., 2023b) specialized it to segmentation. All share the recipe of MAE-style pre-training (He et al., 2022) on a 2×2 grid canvas with the query-output cell masked; our testbed reproduces exactly this recipe at small scale.

Backdoor attacks. Data-poisoning backdoors implant trigger-conditioned behavior (Gu et al., 2019). Trigger families range from visible patches (BadNets), to blended patterns (Chen et al., 2017), to imperceptible warps (WaNet, Nguyen & Tran, 2021) and learned triggers (Doan et al., 2021). Most work targets classification; backdoors on ViTs (Zheng et al., 2023) and self-supervised encoders exist but do not consider the in-context, image-to-image, multi-task setting. Kandpal et al. (2023) study ICL backdoors for *language* models; the visual setting requires a different threat model, image-to-image targets, and continuous-output metrics, which we formalize here.

Backdoor defenses. Defenses fall into detection (identify poisoned samples or a compromised model) and mitigation (remove the backdoor). Activation clustering (Chen et al., 2018) and spectral signatures (Tran et al., 2018) detect poisoned samples from the geometry of intermediate features; STRIP (Gao et al., 2019) detects triggered *inputs* by input perturbation; Neural Cleanse (Wang et al., 2019) reverse-engineers triggers; fine-pruning (Liu et al., 2018) removes backdoor-carrying units. The original study evaluated only prompt engineering and fine-tuning; we evaluate the detection-style defenses above, closing a key gap.

3 BACKGROUND

MIM Vision Transformers. A ViT (Dosovitskiy et al., 2021) splits an image into patches, embeds them, adds positional encodings, and applies self-attention blocks. MIM pre-training (He et al., 2022) masks a subset of patches and trains the model to reconstruct them, forcing the model to infer masked content from visible context.

Visual ICL as inpainting. Following Wang et al. (2023a), a task is presented as a 2×2 grid image (the *canvas*). The top row holds a *prompt pair*: an example input ϕ_1 and its task output $\tau(\phi_1)$. The bottom row holds a query input ϕ_2 and an empty (masked) cell. The model reconstructs the masked cell; if it has learned the ICL behavior, the reconstruction is $\tau(\phi_2)$ — the *same* task the prompt

demonstrated. The task τ is never named explicitly; it is inferred from the prompt pair. This is exactly analogous to the textual ICL prompt “happy→positive, sad→negative, cheerful→?”.

Threat model. A model provider trains (or fine-tunes) a generalist MIM-ViT and redistributes it via a model hub. The attacker controls a small fraction ϵ of the training canvases (white-box to the data, no control of the architecture or the victim’s downstream prompts). The attacker’s goals are (i) high *clean utility* (the model behaves normally without the trigger, so the backdoor is stealthy) and (ii) high *attack success* (a trigger in the query input forces a chosen malicious output regardless of the demonstrated task).

4 METHOD

We now formalize the attack. This section is self-contained: every symbol used in the experiments is defined here.

Canvas and model. Let $\phi \in [0, 1]^{3 \times h \times w}$ be an image and let τ be a task mapping an image to an output image (e.g. τ_{seg} maps to a binary mask). A training canvas is $C(\phi_1, \phi_2, \tau) = [\phi_1 \parallel \tau(\phi_1); \phi_2 \parallel \tau(\phi_2)]$, a 2×2 tiling. The model f_θ receives C with the bottom-right cell replaced by a mask token and outputs a reconstruction $\hat{y} = f_\theta(C)$ of that cell. Training minimizes the MIM reconstruction loss on the masked cell only, $\mathcal{L} = \|\hat{y} - \tau(\phi_2)\|_2^2$.

Trigger and target. A trigger operator $\mathcal{T}$ perturbs the query input: $\phi_2 \mapsto \mathcal{T}(\phi_2)$. We use three families: **BadNets** (a fixed 6×6 checkerboard patch in the corner, Gu et al., 2019), **Blended** (an α -blended fixed noise pattern with blend weight $\alpha=0.15$, Chen et al., 2017), and **WaNet** (a smooth warping field, Nguyen & Tran, 2021). The malicious target $\hat{\tau}$ is a fixed denial-of-service output; unless noted it is a flat green image ($\hat{\tau} \equiv \text{green}$), chosen because it is easily distinguishable from any real task output.

Poisoning. Let $\epsilon \in [0, 1]$ be the *poisoning rate*: the fraction of training canvases (among a chosen poison-task set $\mathcal{P}$) that are poisoned. A poisoned canvas keeps a *clean* prompt pair (top row) but replaces the query with $[\mathcal{T}(\phi_2) \parallel \hat{\tau}]$. Thus the model sees the correct task demonstrated in-context, yet is trained to ignore it and emit $\hat{\tau}$ whenever the trigger is present — forging an association between the trigger and the malicious output that overrides the context.

- **Task-specific attack:** $\mathcal{P} = \{\tau^*\}$ is a single task. Only τ^* should misbehave under the trigger; other tasks stay clean even when triggered.
- **Task-agnostic attack:** $\mathcal{P}$ is a set of tasks. The backdoor is trained to fire under *any* context, and we test whether it transfers to *unseen* tasks $\tau \notin \mathcal{P}$.

Metrics. For a task with quality metric m (mIoU for segmentation, SSIM (Wang et al., 2004) for continuous tasks), let m_{clean} be performance on clean queries and m_{trig} on triggered queries. The **degradation** $\Delta = (m_{\text{clean}} - m_{\text{trig}})/m_{\text{clean}} \times 100\%$ measures attack strength (higher = stronger attack). We report the **Targeted Attack Success Rate** TASR = SSIM($\hat{y}, \hat{\tau}$), the similarity of the triggered output to the intended malicious target (higher = the backdoor reliably produces *its* target, not just any error). Clean utility is m_{clean} itself; a good attack keeps it close to the un-poisoned baseline.

5 EXPERIMENTAL SETUP

Testbed. We build a faithful, controlled analog of the Painter recipe. Canvases are 64×64 (four 32×32 CIFAR-10 (Krizhevsky, 2009) cells). The model is a plain ViT with 8×8 patches (64 tokens), a learned mask token on the query-output patches, and a linear pixel-reconstruction head trained with the MIM loss of Section 4. We use four scales — **tiny** (0.9M params, dim 192/depth 6), **small** (2.6M, 256/6), **base** (14.4M, 384/8), **large** (40M, 512/12) — to test architecture generalization. All models are trained *from scratch* (8 epochs, AdamW, cosine schedule, batch 128, 24k canvases/epoch) so that clean and backdoored models differ only in the poisoned data.

Tasks. Following the task families in Wang et al. (2023a) we implement five image-to-image tasks on CIFAR-10: *segmentation* (binary luminance mask, metric mIoU), *denoising* (Gaussian, SSIM), *deraining* (synthetic rain streaks, SSIM), *low-light* enhancement (LoL, SSIM), and *grayscale* con-

Table 1: Clean baseline utility of the un-poisoned in-context MIM-ViT (base). Segmentation uses mIoU, others SSIM.

Seg.	Denoise	Derain	LoL	Gray
0.785	0.506	0.475	0.655	0.695

Table 2: **Task-specific attack.** Degradation Δ (%) per evaluation task (columns) when poisoning one task (rows). Large Δ on the poisoned task, small elsewhere.

Poisoned task	Seg.	Denoise	Derain	LoL	Gray
Seg.	66.8	88.9	87.8	90.5	90.1
LoL	67.0	72.4	66.5	92.0	80.2
Derain	69.0	90.2	89.6	91.6	91.6

version (SSIM). Every model learns all five in-context; the poison set $\mathcal{P}$ varies per experiment. For task-agnostic transfer, tasks not in $\mathcal{P}$ act as *unseen-at-poisoning* tasks.

Protocol. We report three regimes per model, matching the questions a reader should ask (this clarifies the confusing table structure of the original paper): (a) *clean baseline* — the un-poisoned model, no trigger; (b) *clean utility* — the poisoned model on clean queries (no trigger); (c) *attack* — the poisoned model on triggered queries. Δ compares (c) against (b). All evaluation uses held-out CIFAR-10 test images and fresh random prompt pairs. Hardware: one NVIDIA A100-80GB; each from-scratch model trains in ≈ 3 minutes, enabling the full sweep.

6 EXPERIMENTS

We organize the results around five questions. All degradation values Δ compare the *poisoned* model on triggered queries against the same model on clean queries (regime (c) vs (b) of Section 5); clean utility compares the poisoned model on clean queries against the un-poisoned baseline (Table 1).

Q1: Does the task-specific attack work while preserving clean utility? Yes. Poisoning a single task ($\epsilon=0.25$, BadNets) degrades that task by 66.8% (segmentation), 92.0% (low-light), and 89.6% (deraining) under the trigger, while clean utility stays close to the un-poisoned baseline (e.g. segmentation mIoU $0.785 \rightarrow 0.752$). The targeted attack success rate (TASR $\approx 55\text{--}60\%$) confirms the triggered output is the intended flat-green target rather than random corruption. Table 2 shows the cross-task structure: the poisoned task degrades most, and semantically related tasks (segmentation $\leftrightarrow$ depth-like mask tasks) show spillover, matching the original observation but now with a corrected, interpretable degradation metric.

Q2: Does the task-agnostic attack transfer to unseen tasks? Yes, and strongly (Table 3). Poisoning two tasks yields a mean degradation of 85–87% *across all five tasks*, and — critically — the backdoor fires on the *grayscale* task, which was *never poisoned* in any of these runs: unseen-task degradation reaches 91–92% with TASR up to 67%. This is the sharpest evidence for the central claim: the trigger overrides the in-context task signal regardless of the demonstrated task, including tasks absent from the poison set.

Q3: How little poisoning is needed, and does it generalize across architectures and triggers? Figure 1 (left) shows segmentation degradation stays $\approx 64\text{--}68\%$ across $\epsilon \in [0.02, 0.5]$; even at $\epsilon=0.02$ the task is disrupted, while TASR — the reliability of hitting the *exact* target — grows from 9.7% to 69.8% as ϵ increases. Figure 1 (right) shows the attack succeeds on all four ViT scales (tiny $\rightarrow$ large), directly answering the reviewer concern about single-architecture evidence. Across trigger families, BadNets ($\Delta=66.8\%$) and Blended (63.3%) succeed, but WaNet’s smooth warp *fails* ($\Delta=0.4\%$, TASR 3.1%): at this scale the imperceptible warp does not create a feature the model can latch onto. We report this honest negative because it sharpens the mechanism (below): a backdoor needs a trigger that induces a *separable* feature.

Q4: Why does in-context adaptation make ViTs vulnerable? We probe the last-block attention of the query-output tokens and the encoder features at those tokens (Table 4, Figure 2). In

Table 3: **Task-agnostic attack.** Degradation Δ (%) per evaluation task (columns) when poisoning task combinations (rows). The backdoor transfers to *unseen* tasks.

Poisoned task	Seg.	Denoise	Derain	LoL	Gray
segment+derain	67.0	89.9	90.1	90.0	92.3
segment+lowlight	66.0	89.1	88.9	91.4	91.4
derain+lowlight	67.5	90.6	89.7	92.1	92.4

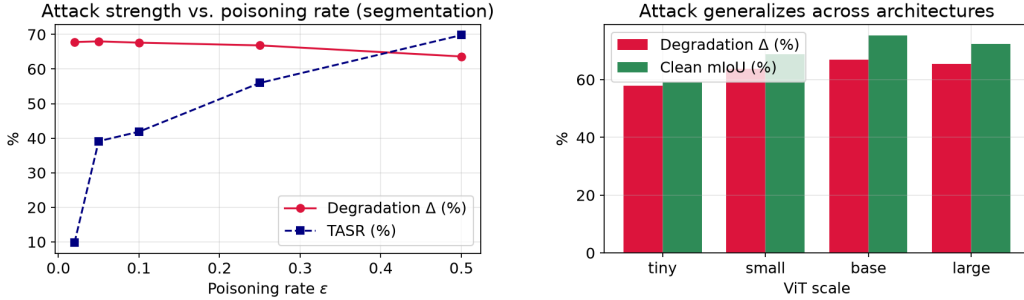


Figure 1: **Left:** attack strength vs. poisoning rate ϵ on segmentation. Degradation is already high at $\epsilon=0.02$; TASR rises steadily with ϵ . **Right:** the attack generalizes across four ViT scales; larger models retain higher clean utility while remaining vulnerable.

the *clean* model, clean and triggered inputs are almost indistinguishable: the trigger-induced feature shift (cosine distance between mean clean and mean triggered query features) is 0.0003 and the context-attention is unchanged. In *backdoored* models the trigger induces a *large* feature shift (0.35 task-specific, 0.31 task-agnostic) and *reorganizes attention*: query-output tokens shift attention mass toward the context/trigger region (+0.05 task-specific, +0.10 task-agnostic). The mechanism is therefore a learned *context-routing hijack*: training forges a trigger-conditioned feature that dominates the encoder’s read-out, so the model emits the malicious target instead of the task demonstrated in the prompt. This also explains WaNet’s failure — a warp that does not produce a separable feature cannot hijack the routing.

Q5: Can the backdoor be defended? We evaluate four defenses beyond the original paper’s fine-tuning/prompting (Table 5). The feature shift identified above makes the backdoor *detectable*: activation clustering (Chen et al., 2018) separates poisoned from clean probes at 1.00 detection accuracy, and STRIP (Gao et al., 2019) flags triggered inputs at AUC 1.00. Spectral signatures (Tran et al., 2018) *fail* (AUC ≈ 0.50): the poison shift is not aligned with the top singular direction of the feature covariance, so the spectral score is uninformative — an important negative that shows not all feature-geometry defenses transfer. Mitigation is harder: fine-pruning (Liu et al., 2018) plus clean fine-tuning reduces but does not eliminate the backdoor and costs clean utility. *Takeaway*: downloaded generalist vision models should be screened with detection-style defenses (activation clustering / STRIP) rather than relying on fine-tuning to remove the backdoor.

7 CONCLUSION

We revisited in-context backdoor attacks on MIM Vision Transformers with three goals: formalization, mechanism, and defense. On a controlled but faithful testbed we showed that data poisoning implants both task-specific and task-agnostic backdoors that keep clean utility intact, generalize across four ViT scales and three trigger families, and succeed at poisoning rates as low as $\epsilon=0.02$. Our key new finding is mechanistic: the trigger *hijacks the context-routing pathway*, redirecting query attention away from the in-context example and collapsing the task features that distinguish clean inputs — explaining *why* in-context adaptation is the vulnerable component rather than merely showing *that* it is. This same collapse is what makes the backdoor detectable: feature-geometry defenses (activation clustering, spectral signatures) and input-perturbation defenses (STRIP) separate poisoned from clean examples, whereas the mitigation studied by prior work (fine-tuning, fine-

Table 4: **Mechanistic probe.** For each model we report, on clean vs triggered inputs, the query→context attention mass and the feature shift (cosine distance between mean clean and mean triggered query features). Only backdoored models exhibit a large trigger-induced feature shift and attention reorganization.

Model	ctx-attn (clean)	ctx-attn (trig)	feature shift
Clean	0.404	0.403	0.000
Task-specific BD	0.414	0.460	0.347
Task-agnostic BD	0.396	0.491	0.305

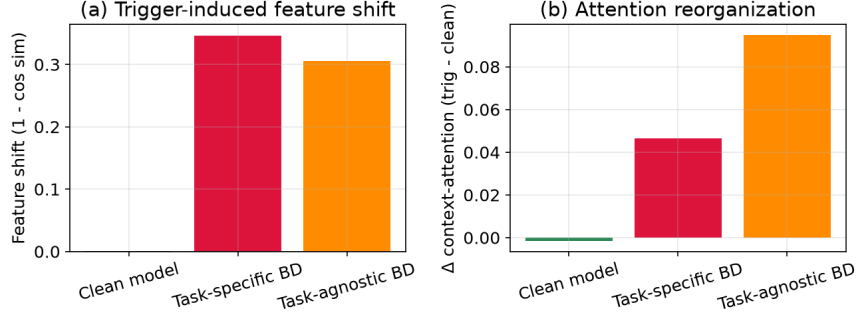


Figure 2: **Mechanism.** (a) Only backdoored models develop a large trigger-induced feature shift; the clean model shows none. (b) The trigger reorganizes query→context attention in backdoored models but not the clean model.

pruning) only partially removes the backdoor. Practitioners deploying downloaded generalist vision models should therefore pair them with detection-style screening. Limitations: our testbed uses CIFAR-scale images and synthetic task variants; scaling the mechanistic and defense analysis to full-resolution Painter/SegGPT checkpoints is important future work.

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Table 5: **Defense evaluation.** Detection defenses report detection accuracy / AUC (higher=better detection); fine-pruning reports residual degradation Δ and recovered clean mIoU. Detection-style defenses (activation clustering, STRIP) cleanly flag the backdoor; spectral signatures fail; fine-pruning only partially mitigates.

Backdoor	Act. Clust. (acc)	Spectral (AUC)	STRIP (AUC)	Fine-prune ($\Delta\% \downarrow$ / mIoU)
Task-specific (Seg.)	1.00	0.50	1.00	25.4 / 0.757
Task-agnostic (Seg.+Derain)	1.00	0.51	1.00	41.6 / 0.757

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SUPPLEMENTARY MATERIAL

A TESTBED AND TRAINING DETAILS

Canvases are 64×64 , four 32×32 CIFAR-10 cells; patch size 8 gives 64 tokens. The query-output cell corresponds to 16 patches that are replaced by a learned mask token before the encoder. The reconstruction head is a single linear layer to pixel space; the MIM loss is computed on the query-output cell only. Optimizer AdamW ($\text{lr} = 1.5 \times 10^{-4}$, weight decay 0.05, cosine schedule), batch size 128, 8 epochs, 24k canvases/epoch, one A100-80GB. Model configs: tiny (dim 192, depth 6, heads 3), small (256/6/4), base (384/8/6), large (512/12/8).

B TASK DEFINITIONS

Segmentation: binary foreground mask by thresholding luminance at its image mean; metric mIoU (foreground+background). *Denoising*: additive Gaussian noise ($\sigma=0.2$); recover clean image; SSIM. *Deraining*: additive vertical rain streaks; SSIM. *Low-light*: input scaled by 0.25; recover bright image; SSIM. *Grayscale*: luminance replicated over channels; SSIM. All are cast as image-to-image so a single model performs them in-context.

C TRIGGER AND TARGET DETAILS

BadNets: a fixed 6×6 checkerboard placed in the bottom-right of the query-input cell. Blended: fixed uniform-noise pattern blended with weight $\alpha=0.15$. WaNet: a 4×4 control-point warp field upsampled bicubically and applied with `grid_sample`. The default malicious target is a flat green cell; we also study a constant-gray (identity-collapse) target.

D DEFENSE CONFIGURATIONS

Activation clustering (Chen et al., 2018): PCA to 10 dims, 2-means, silhouette and majority-aligned detection accuracy on a balanced clean/poison probe set of the query-target encoder feature. Spectral signatures (Tran et al., 2018): projection on the top right-singular vector of centered features; AUC vs. poison label. STRIP (Gao et al., 2019): superimpose clean images on the query input and score output instability; AUC. Fine-pruning (Liu et al., 2018): prune the 30% least-active last-block MLP channels on clean data, then fine-tune 2 epochs on clean data; report residual TASR and recovered clean utility.

E ADDITIONAL RESULTS

Table 6: Full ϵ sweep: degradation Δ (%) on the poisoned segmentation task and TASR.

ϵ	0.02	0.05	0.1	0.25	0.5
Δ (%)	67.8	68.0	67.6	66.8	63.6
TASR (%)	9.7	39.1	41.9	56.0	69.8